

Gavin Loughrie

04/25/25

Data Science Final Report

Data Science on NBA Odds

This final project was all about a dataset named NBA Odds Data. This dataset has information about every basketball game in the NBA from the 2008 to 2023 seasons. This spreadsheet is very large, at over 37000 rows of data, the different columns are split up into different stats that include: game date, team names, home or away status, scores, betting odds (Money Line, spread, and totals), and second-half statistics are among the 12 columns of data it contains. This dataset highlights game results as well as the relationship between betting odds and team performance over the fifteen-year period. Here is a visualization of this spreadsheet:

Figure 1.1 NBA Odds Spreadsheet

oddsData											
date	season	team	home/visitor	opponent	score	opponentScore	moneyLine	opponentMoneyLine	total	spread	secondHalfTotal
2007-10-30	2008	Utah	@	Golden State	117	96	100	-120	212	1	105.5
2007-10-30	2008	LA Lakers	vs	Houston	93	95	190	-230	199	5	99
2007-10-30	2008	Houston	@	LA Lakers	95	93	-230	190	199	-5	99
2007-10-30	2008	San Antonio	vs	Portland	106	97	-1400	900	189.5	-13	95
2007-10-30	2008	Portland	@	San Antonio	97	106	900	-1400	189.5	13	95
2007-10-30	2008	Golden State	vs	Utah	96	117	-120	100	212	-1	105.5
2007-10-31	2008	New Jersey	vs	Chicago	112	103	-125	105	186	-1.5	94
2007-10-31	2008	Dallas	@	Cleveland	92	74	-140	120	184	-2.5	91.5
2007-10-31	2008	Cleveland	vs	Dallas	74	92	120	-140	184	2.5	91.5
2007-10-31	2008	Seattle	@	Denver	103	120	625	-825	217.5	11.5	107.5
2007-10-31	2008	Washington	@	Indiana	110	119	-125	105	203.5	-1.5	105
2007-10-31	2008	San Antonio	@	Memphis	104	101	-250	210	201	-6.5	99.5
2007-10-31	2008	Orlando	vs	Milwaukee	102	83	-295	245	197	-6.5	97
2007-10-31	2008	Chicago	@	New Jersey	103	112	105	-125	186	1.5	94
2007-10-31	2008	Sacramento	@	New Orleans	90	104	425	-525	190.5	9.5	94
2007-10-31	2008	Milwaukee	@	Orlando	83	102	245	-295	197	6.5	97
2007-10-31	2008	Toronto	vs	Philadelphia	106	97	-305	255	191	-6.5	96.5
2007-10-31	2008	New Orleans	vs	Sacramento	104	90	-525	425	190.5	-9.5	94
2007-10-31	2008	Memphis	vs	San Antonio	101	104	210	-250	201	6.5	99.5
2007-10-31	2008	Denver	vs	Seattle	120	103	-825	625	217.5	-11.5	107.5
2007-10-31	2008	Philadelphia	@	Toronto	97	106	255	-305	191	6.5	96.5
2007-10-31	2008	Indiana	vs	Washington	119	110	105	-125	203.5	1.5	105
2007-11-01	2008	Miami	vs	Detroit	80	91	160	-190	178.5	4	89.5
2007-11-01	2008	Utah	vs	Houston	95	106	-160	140	193	-3	98.5
2007-11-01	2008	Detroit	@	Miami	91	80	-190	160	178.5	-4	89.5
2007-11-01	2008	Seattle	vs	Phoenix	99	106	575	-725	224.5	12	112.5
2007-11-01	2008	Phoenix	@	Seattle	106	99	-725	575	224.5	-12	112.5
2007-11-01	2008	Houston	@	Utah	106	95	140	-160	193	3	98.5

There is a lot of data crammed into this spreadsheet that can be examined and used for predictions in the future. A number of audiences, including sports analysts, bettors, and machine learning developers, can benefit from knowing how betting odds relate to game outcomes. This analysis looks at how betting spreads have changed over time, how accurate Money Line odds

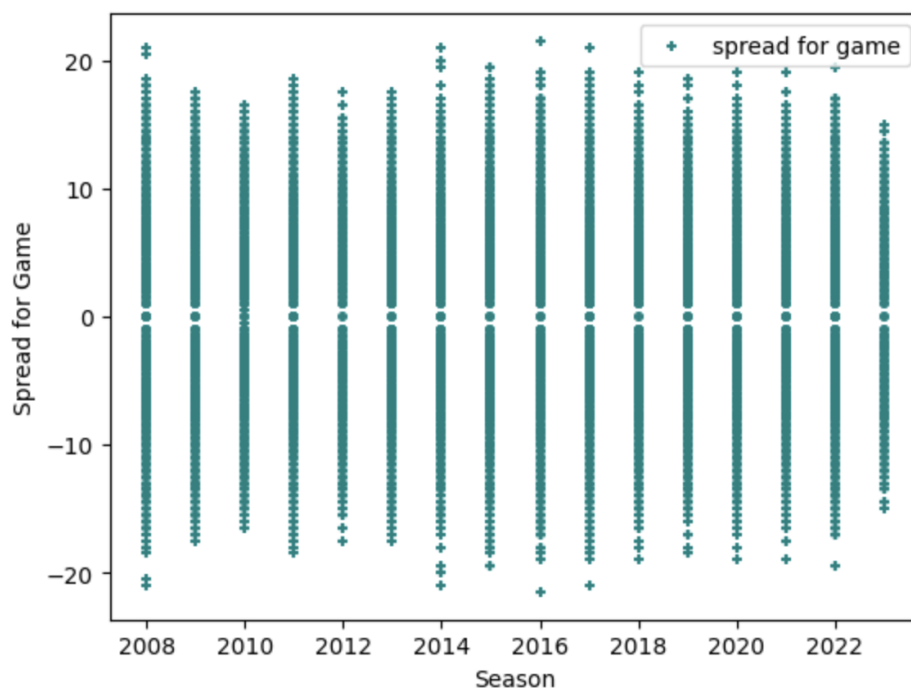
are at picking winners, and whether machine learning can be used to predict game outcomes based on pregame information. Sports bettors looking for a statistical advantage and machine learning professionals wishing to apply predictive models to actual sports data may find these insights useful.

My passion for basketball and my curiosity in sports betting, especially how Vegas determines its odds, led me to select this dataset for my final project. I've been an avid NBA fan for years, and I've always found the game's strategy and unpredictable environment to be fascinating. Because of this, the NBA Odds Data was a perfect match. The chance to investigate how betting odds, such as the Money Line, spreads, and totals, are calculated and how closely they correspond to actual game results intrigued my interest even more. I've frequently thought about how sportsbooks create odds that accurately reflect a team's chances of winning while also drawing bets based on factors like team matchups, injuries, and past performance. If you are not familiar with the different terms, like Money Line or spread, they are betting terms. Money Line refers to a bet that means that the team will win, and a negative number means that they are favored, and the higher the negative, the more favored they are to win. Spread refers to how much the team will win by. For example, say the Celtics spread against the Cavaliers is -1.5. If the Celtics win by 2 points, then the bet would hit, and the person would win money. However, if the spread is +2.5, then all the Celtics would have to do is lose by no more than 2 points, and the bet would hit.

My first question for this project was looking at how the spread odds have changed throughout the NBA season from 2008-2023. This question is very important because looking at how spreads have shifted from 2008 to 2023 can reveal broader trends in the NBA, like changes in team dominance, scoring patterns, rule changes (like pace of play or emphasis of the three-point line), and how Vegas responds to those changes. By analyzing this, you're essentially

looking into how sportsbooks adjust to league dynamics and betting behavior. This helps bettors identify whether certain seasons were more unpredictable and gives analysts a window into how statistical expectations have evolved. To start, I made a scatter plot to plot the season on the x-axis and the spreads on the y-axis. I then calculated a Pearson's r correlation score to show the correlation between the time of the season and the spread. The data was already processed, so I did not need to process anything specifically. Here is a visualization of the graph I made:

Figure 1.2 Scatterplot of Spreads throughout NBA Seasons



from the graph, you can tell that most spreads fall between about -10 and +10 points, meaning that in most games, no team was expected to win or lose by more than 10 points. However, there are some outliers each year, where the spread is much larger (around +20 or -20), showing games where one team was heavily favored. The spread values seem fairly stable across the years, suggesting that while matchups might vary, the way sportsbooks set spreads for NBA

games has remained consistent over time. The correlation score ended up being -4 , and here is the exact number:

Figure 1.3 Pearson R correlation Score of Figure 1.2

```
: corr = scipy.stats.pearsonr(season, spread)
print(corr)
print(corr[0])
correlation_score = corr [0]
print(correlation_score)

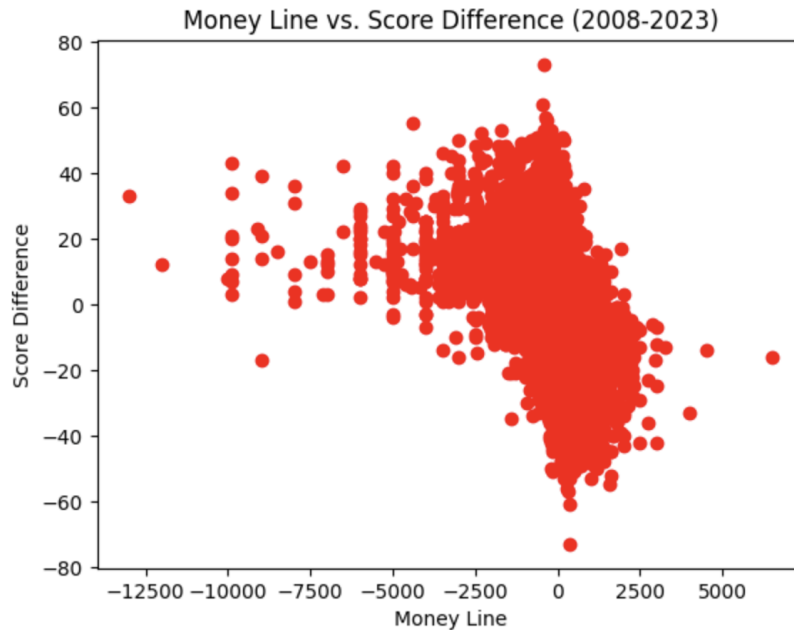
PearsonRResult(statistic=np.float64(-4.336808689942018e-19), pvalue=np.float64(0.9999999999999993))
-4.336808689942018e-19
-4.336808689942018e-19

: print(correlation_score)
-4.336808689942018e-19
```

The score being that low shows that there is no correlation between the season and the spread. I suspect that because the spreads were very similar throughout the different seasons, and so the correlation would be that low. I found this surprising because to me, the NBA gets more and more talented every year, and so the scores differential should be higher. I wonder in the coming years if we will see more teams beating other teams by more and having more inconsistent trends with the spreads.

For my second question, I wanted to look at how often the favored team wins and the score difference of those games. I used a scatter plot and put the different Money Line odds on the x-axis and on the y-axis. I created a variable by taking the score minus the opponent score to show the outcome of the games (score differential). I calculated a Pearsons r correlation score to show how well they are connected. This was very important to analyze because, if the favored team wins more often then it would show that sportsbooks put out accurate predictions and they are not being unfair. On the other hand, if underdogs are winning more often then it would show how unpredictable the NBA is. Including the score differential will give insight as to whether the teams won by a large margin or if they were close.

Figure 1.4 Scatter plot of Money Line odds and Score Differential



This graph displays an expected trend that the favored teams will usually win and by a large amount, and vice versa that the “underdogs” usually lose, and by a decent number of points. There is some spread, meaning that not every favorite wins and not every underdog loses, but the overall pattern supports the idea that the Money Line is a decent predictor of who will win.

Figure 1.5 Pearsons R correlation score of Figure 1.4

```
corr = scipy.stats.pearsonr(money_line, score_difference)
print(corr)
print(corr[0])
correlation_score = corr [0]
print(correlation_score)
```

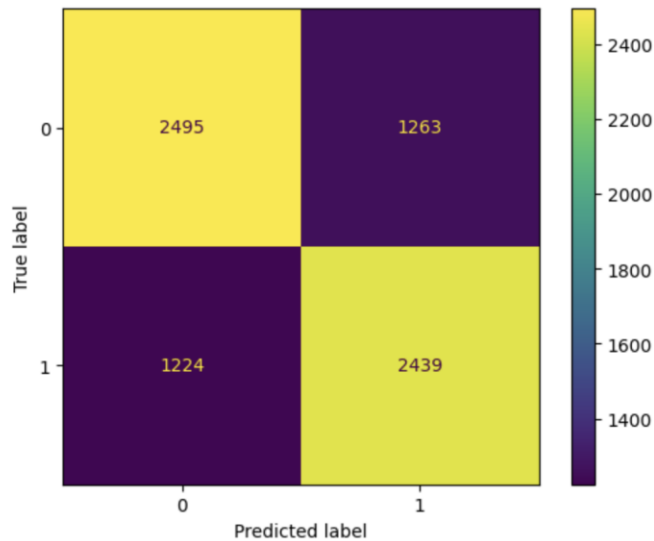
```
PearsonRResult(statistic=np.float64(-0.39335288766190873), pvalue=np.float64(0.0))
-0.39335288766190873
-0.39335288766190873
```

The Pearsons r correlation score being -0.3 between the Money Line and score difference highlights the weak negative relationship. This means that as the odds for the Money Line get worse, the score difference margin tends to get smaller. This was not surprising to me, it just makes sense that as the closer the game/the harder it is to predict the game, that the score

difference would be closer. This supports that idea that the sportsbook lines are accurate and consistent.

For my final question I wanted to know if my machine learning model could predict the outcome of a game based on certain categories such as Money Line, opponent Money Line, spread, total, and Second Half Total. These categories are obviously used by the different sports books to create their odds, so it should be the perfect data to plug into my model to help me predict the outcomes. If my model can successfully predict these games, it could be very valuable for the bettors, analysts, and even potentially used for scouting objectives. The steps I took to accomplish this is as follows: I made a correlation matrix to see what categories are most compatible with my output category. I created my variables and train-test split. Then, using KNN (K Nearest Neighbors) classification to create a confusion matrix to show the model. KNN is a machine learning algorithm that predicts the outcome of a data point based on the outcomes of the “K” most similar data points nearby. It looks at the closest examples based on the features you provide, and uses majority voting to decide the prediction. The only data I had to process was to create a new output value by taking the columns Score and Opponent Score, and creating a new integer when Score is greater than the opponent score naming it win.

Figure 1.6 Confusion Matrix of NBA Game Predictions



I was very pleased with how this confusion matrix turned out. The two yellow boxes represent the number of times my model predicted a team to win and a team to lose. The purple box represents the number of times my model predicted a team to win, and they lost and vice versa. With that in mind, it tells me that my model is right around 2/3 or 67% of the time. Being able to be that accurate with just the odds is very good and it is a little interesting. I am kind of surprised that it was able to do so well, but that is the power of machine learning and how smart the model became after I used hyperparameter tuning. To go deeper, hyperparameter tuning is the process of adjusting the model's settings to improve its performance. I changed the number of neighbors used in the K-nearest neighbors algorithm, which tells the model to use a different number of surrounding games it should consider when making a prediction. This allowed me to get the best output and generate the most accurate matrix.

I learned a lot about how betting odds correspond with game results from my study of the NBA Odds dataset from 2008 to 2023. With little to no connection to the season, I discovered that point spreads have stayed relatively constant over time, indicating that sportsbooks have continued to set expectations consistently. A weak but reasonable correlation between Money

Line odds and score differentials was also revealed by my research, indicating that even though favored teams typically win, upsets are still possible. Lastly, my machine learning model demonstrated influencing potential for future predictive tools in sports analytics with a prediction accuracy of about 67% using only betting-related inputs. All things considered, this project improved my understanding of the analysis of sports betting data, and it emphasized how data science and basketball, two things I'm deeply invested in, and how they intersect. The significance of careful data preparation, visualization, and model tuning to generate significant insights was also emphasized by this entire experience.

Works Cited

Treasure, C. (2024, January 5). *NBA Odds Data*. Kaggle.

<https://www.kaggle.com/datasets/christophertreasure/nba-odds-data>

NBA point spread betting guide - understand NBA spread bets. Free Expert Sports Picks. (n.d.).

<https://onlinebetting.com/basketball/nba/point-spread>